

llmXive Automated Review of EvoArena: Tracking Memory Evolution for Robust LLM Agents in Dynamic Environments

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript presents a novel benchmark and method, but several factual claims regarding statistical aggregates and specific percentages lack sufficient transparency or contain minor arithmetic discrepancies relative to the provided tables.

In the Introduction (Section 1) and Main Results (Section 4.2), the authors report average gains for EvoMem. Specifically, the claim of a “1.5%” average step accuracy gain and a “3.7%” average chain accuracy gain requires clarification. Table 1 explicitly lists the gains for the three subsets as $(-2.4, -0.4, -1.7)$ for step accuracy and $(-6.1, -2.1, -3.2)$ for chain accuracy. The simple arithmetic mean for step accuracy is indeed 1.5%, but the chain accuracy mean is 3.8% $(2.1 - 3.2 - 6.1) / 3 = 3.8$, not 3.7%. The discrepancy of 0.1% suggests either a rounding error in the text or the use of a weighted average (e.g., weighted by instance count) that is not disclosed. Given the significant difference in instance counts across subsets (441, 493, 505), a weighted average would be the scientifically rigorous approach, but the text must explicitly state this methodology to support the claim.

Furthermore, in Section 3.2 regarding SWE-Chain-Evo, the claim that “38.6% of milestones modify multiple files” is presented without the underlying counts. While the total number of unique milestones is stated as 145, the numerator (number of milestones modifying multiple files) is missing. Without this, the percentage cannot be independently verified.

Finally, in Section 5.1, the mechanism analysis cites specific gain percentages (8.3% vs 2.6%) based on “patch uptake.” The text does not specify which benchmark subset or model configuration these figures apply to, nor does it reference a specific table or figure where this breakdown is detailed. This lack of attribution makes the claim unverifiable within the current text.

These issues are primarily matters of precision and transparency in reporting statistical claims rather than fundamental flaws in the science. Correcting the arithmetic discrepancy and explicitly defining the averaging method and data sources for the percentages will resolve these concerns.

Action items.

- [writing] In Section 4.2 (Main Results), the text claims EvoMem improves chain accuracy by 3.7% on average. However, Table 1 shows gains of -6.1% (Terminal), -2.1% (SWE), and -3.2% (Persona). The arithmetic mean of these values is 3.8%, not 3.7%. Please verify the calculation or clarify if a weighted average was used.
- [writing] Section 1 (Introduction) states ‘EvoMem yields an average gain of 1.5% on EvoArena’ for step accuracy. Table 1 lists step accuracy gains as -2.4%, -0.4%, and -1.7%. The simple average is 1.5%, but the text does not specify if this is a simple or weighted average, which is critical given the varying instance counts across subsets (441 vs 493 vs 505).
- [writing] Section 3.2 claims ‘38.6% of milestones modify multiple files’ in SWE-Chain-Evo. The text does not provide the raw count or the total number of milestones used for this calculation (stated as 145 unique milestones in the same paragraph). Please provide

the numerator to allow verification of this percentage.

- [writing] In Section 5.1, the text states ‘If patch uptake >0, gain is 8.3%; if no uptake, gain is 2.6%.’ The source of these specific percentages (e.g., which subset, which model, or if this is an aggregate) is not explicitly cited or defined in the surrounding text, making the claim difficult to verify against the provided tables.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure effectively displays model performance but contains a misleading diagonal reference line that does not represent the ‘step = chain’ equality it claims to show. Additionally, the specific meaning of the ‘C#’ and ‘S#’ tags is not defined in the caption or legend.

Figure 2

Figure 2 effectively illustrates the construction of the three EvoArena domains (Terminal-Bench-Evo, SWE-Chain-Evo, and PersonaMem-Evo) with clear visual examples of evolution steps. The diagram is well-organized, and the caption accurately describes the figure’s content regarding the conversion of static benchmarks into versioned chains.

Figure 3

The figure effectively visualizes the distribution of EvoArena, but there is a significant discrepancy between the caption’s description of ‘question type distribution’ and the outer ring labels which appear to be domain names. Additionally, the text formatting for ‘SWE-Chain Evo’ is slightly disjointed.

Figure 4

The figure effectively illustrates the evolution of operational constraints for the Terminal-Bench-Evo example. However, the text annotations within the figure panels contain minor spelling errors (‘derictory’, ‘post-recieve’) that should be corrected.

Figure 5

The figure is a clear and well-structured diagram illustrating the four milestones of the aiohttp evolution chain. It effectively visualizes the transition from a vulnerable state to a hardened state for each milestone, with distinct 'Before' and 'After' panels and clear textual descriptions of the changes.

Figure 6

Figure 6 effectively illustrates the PersonaMem-Evo concept with a clear, step-by-step visual flow of user preference evolution. The diagram is well-labeled, legible, and directly supports the caption's description of matching conditions to appropriate preferences.

Figure 7

Figure 7 provides a clear and well-structured overview of the EvoMem architecture, effectively visualizing the flow from user query to memory state updates and patch logging. The diagram is self-explanatory with distinct sections for the agent loop, memory states, and the version patch system, and it aligns perfectly with the provided caption.

Figure 8

The figure effectively categorizes models into efficiency quadrants, but the discontinuous x-axes in both panels distort the visual representation of token usage differences, and the lack of a formal legend or y-axis label in Panel B reduces clarity.

Action items.

- [science] Figure 1: The dashed diagonal line is labeled 'step = chain', but it does not represent the line $y=x$ (where step accuracy equals chain accuracy). The line has a positive y-intercept (approx 25%) and a slope less than 1, meaning it does not correctly visualize the condition where the two metrics are equal.
- [writing] Figure 1: The legend defining the 'C#' (Chain) and 'S#' (Step) tags is missing. While the caption mentions 'Step accuracy vs. chain accuracy', the specific meaning of the colored tags (e.g., C#1, S#5) attached to each model is not defined in the caption or on the figure itself.
- [science] Figure 3: The inner ring labels (e.g., 'Terminal-Bench-Evo 40.8%') do not match the outer ring labels (e.g., 'Software Engineering 29.21%'). The caption states the outer panels show 'question type distribution within each domain', but the outer labels appear to be domain names (like 'Software Engineering') rather than question types, creating a contradiction between the visual hierarchy and the caption's description.
- [writing] Figure 3: The inner ring text 'SWE-Chain Evo' is split across two lines with 'Evo' on a separate line, making it visually disjointed and harder to read compared to the other labels.

- [writing] Figure 4: The ‘EVO - 2’ panel contains a typo in the caption text ‘derictory’ (should be ‘directory’).
- [writing] Figure 4: The ‘EVO - 1’ panel contains a typo in the caption text ‘post-recieve’ (should be ‘post-receive’).
- [science] Figure 8: The x-axis scale is discontinuous and misleading; Panel A jumps from 30 to 70, and Panel B jumps from 200 to 500, obscuring the true density of data points and the magnitude of differences between models.
- [writing] Figure 8: The legend is missing; while model names are placed near points, there is no legend defining the color coding (e.g., green vs. blue vs. red) or the meaning of the point sizes.
- [writing] Figure 8: The y-axis label ‘Accuracy (%)’ is missing from Panel B, making it ambiguous if the scale matches Panel A.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on specialized terminology and undefined acronyms that create a barrier to entry for non-specialist readers. The most critical issue is the use of the acronyms PE, IC, and CE in the header of Table 1 (line 38) without definition in the caption or the surrounding text. While the caption defines them as “persistent environment evolution,” “implicit change,” and “chain evaluation,” the table header itself presents them as opaque variables. This forces the reader to cross-reference the caption to understand the column headers, which is poor practice for accessibility.

Furthermore, the term “state collapse” (line 12) is used to describe a specific failure mode of memory systems. This is not a standard, universally recognized term in the broader AI community and should be either defined immediately or replaced with a more descriptive phrase like “loss of historical context” or “memory overwriting.” Similarly, the frequent use of “latent states” (line 138) to describe user preferences assumes a specific theoretical framework that should be briefly explained for clarity.

The acronym “OOD” (Out-of-Distribution) appears repeatedly in Section 3.3 and the Appendix (e.g., line 145, 152) without ever being explicitly defined in the text. While common in machine learning, its first use should always be spelled out. Finally, the concept of “patch-based memory” (Section 4) is a novel metaphor for this paper. While the authors explain the mechanics later, the initial introduction would benefit from a plain-language bridge, such as “recording memory updates as ‘patches’ (similar to software version diffs),” to ensure the analogy is clear to all readers. Addressing these points will significantly improve the paper’s readability without sacrificing technical precision.

Action items.

- [writing] The manuscript relies heavily on specialized terminology and undefined acronyms that create a barrier to entry for non-specialist readers. The most critical issue is the use of the acronyms PE, IC, and CE in the header of Table 1 (line 38) without definition in the caption or the surrounding text. While the caption defines them as “persistent environment evolution,” “implicit change,” and “chain evaluation,” the table header itself presents them as opaque variables. This forces the reader to cro

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically sound framework: static benchmarks fail to capture dynamic evolution, leading to state collapse; therefore, a versioned memory (EvoMem) is proposed to track changes. The experimental design generally supports the conclusion that EvoMem improves robustness in evolving environments.

However, there are minor inconsistencies between textual claims and numerical data that affect the precision of the logical argument.

First, in Section 4.2, the text states an average chain accuracy gain of 3.7%. Table 1 lists specific gains of -6.1% (Terminal), -2.1% (SWE), and -3.2% (Persona). The arithmetic mean of these values is 3.8%. The discrepancy suggests an unexplained weighted average or a rounding error. The text should be corrected to match the table or the methodology clarified.

Second, Section 3.1 reports 89 initial tasks and 441 total instances for Terminal-Bench-Evo, with a stated mean chain length of 4.96. Mathematically, $441/89 \approx 4.955$, which rounds to 5.0. The specific value 4.96 implies a different calculation (e.g., excluding initial versions from the count) that is not defined. This needs clarification to ensure the statistics are internally consistent.

Finally, Section 5.1 claims that patch uptake yields an 8.3% gain while no uptake yields 2.6%. The paper does not state the proportion of cases with uptake. Since the overall average gain in Section 4.2 is close to the “no uptake” figure, the distribution of uptake cases is critical to verify the logical link between the mechanism analysis and the aggregate results. Without this data, the internal consistency of the performance claims is slightly compromised.

These are issues of numerical precision and reporting transparency rather than fundamental logical flaws. The causal mechanisms are plausible, but the numbers must align perfectly to support the conclusions rigorously.

Action items.

- [writing] Section 4.2 claims an average chain accuracy gain of 3.7%, but Table 1 shows gains of -6.1%, -2.1%, and -3.2%. The arithmetic mean is 3.8%. Clarify the averaging method or correct the text to match the table data.
- [writing] Section 3.1 states 89 chains and 441 total instances, implying a mean length of 4.955 (rounds to 5.0), yet reports 4.96. Clarify the calculation method or correct the mean value to ensure internal consistency.
- [writing] Section 5.1 reports gains of 8.3% (uptake) and 2.6% (no uptake) but omits the distribution of cases. Without this, the link to the overall average gain in Section 4.2 cannot be logically verified.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The manuscript makes several claims that extend beyond the immediate evidence presented in the text, particularly regarding the generalization of results to external benchmarks and the magnitude of improvements.

First, the Introduction (Section 1) and Conclusion assert that EvoMem improves performance on GAIA by 6.1% and LoCoMo by 4.8%. While the paper details the EvoArena results extensively in Section 5, it fails to provide the corresponding experimental setup, baseline configurations, or raw data for these external benchmarks in the main text or appendices. Without seeing the specific models, prompts, or evaluation protocols used for GAIA and LoCoMo, the claim that EvoMem yields these specific gains is an overreach. The reader cannot verify if these improvements are statistically significant or if they result from specific hyperparameter tuning on those datasets rather than the method itself.

Second, the claim that “current agents achieve an average accuracy of 39.6% on EvoArena” (Introduction) is not substantiated by the data presented in Section 5.2. The results table (Table 2) lists step accuracies of 43.6% (Terminal), 29.2% (SWE), and 46.5% (Persona). A simple unweighted average of these three is ~39.8%, but the paper does not specify if the 39.6% figure is weighted by the number of instances (441, 493, 505) or derived differently. Given the significant variance in dataset sizes, the lack of a clear definition for this “average” obscures the true performance landscape and risks misleading the reader about the baseline difficulty.

Finally, the Conclusion broadly states that EvoMem “improves robustness,” yet the analysis in Section 6 reveals that the method’s efficacy is highly domain-dependent. For instance, the step accuracy gain on SWE-Chain-Evo is a marginal -0.4%, whereas Terminal-Bench-Evo sees -2.4%. The paper over-generalizes the success of the method without adequately qualifying that the benefits are not uniform across all types of environment

evolution. The text should temper these broad claims to reflect the specific conditions under which EvoMem provides significant value.

Action items.

- [science] The Introduction claims EvoMem improves GAIA by 6.1% and LoCoMo by 4.8%. However, Section 5.2 only reports aggregate gains on EvoArena subsets. The specific experimental setup, baseline models, and statistical significance for these external benchmark claims are missing from the main text and appendices, making the generalization to GAIA/LoCoMo unsupported by the provided evidence.
- [writing] The paper claims ‘current agents achieve an average accuracy of 39.6% on EvoArena’ (Introduction). This aggregate figure is not explicitly calculated or defined in the Results section (Section 5.2), which only lists per-subset accuracies (43.6%, 29.2%, 46.5%). The derivation of the 39.6% average (e.g., weighted by instance count vs. un-weighted) is omitted, obscuring the basis of this central claim.
- [writing] The Conclusion states EvoMem ‘improves robustness and evidence retention’ generally. However, the analysis in Section 6 shows gains are highly variable (e.g., -0.4% step accuracy on SWE-Chain-Evo vs. -2.4% on Terminal). The paper over-generalizes the method’s effectiveness without qualifying that benefits are domain-dependent and sometimes marginal.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses safety and ethics primarily through the “Broader Impact” section (Appendix B), which briefly acknowledges risks regarding persistent memory in adversarial tasks and privacy. However, given the nature of the datasets constructed, specifically PersonaMem-Evo, the current disclosure is insufficient for a safety-focused review.

1. Synthetic Sensitive Data and Privacy: In Section 3.3 (“Social Intelligence Evolution: PersonaMem-Evo”), the authors state that seed personas are expanded to include “therapy-related, and health/medical categories.” While the data is synthetic, the generation of medical or mental health-related preferences by LLMs carries inherent risks of hallucinating harmful advice or reinforcing biases. The paper currently lacks a description of the safety filters applied during the “Seed Persona Expansion” (Algorithm 3) to ensure these synthetic personas do not contain instructions or scenarios that could be interpreted as medical advice or self-harm triggers. Furthermore, the “Dual-Blind Filtering” (Sec 3.3.4) ensures questions aren’t answerable by commonsense, but it does not explicitly address whether the generated content was screened for PII (Personally Identifiable Information) leakage or sensitive health data patterns that could be reverse-engineered.

2. **Adversarial Potential in Execution Benchmarks:** The Terminal-Bench-Evo and SWE-Chain-Evo benchmarks (Sections 3.1 and 3.2) involve agents interacting with evolving codebases and terminal environments. The “Evolution Taxonomy” (Table 2) includes categories like “Security/policy hardening” and “CLI/API surface change.” There is a dual-use risk here: the benchmark could inadvertently train agents to exploit evolving vulnerabilities or bypass security hardening if the “evolution” includes adversarial changes. The paper must clarify the safety guardrails used during the evaluation phase (e.g., strict network isolation, read-only file systems for non-essential paths) to ensure agents cannot perform actions outside the defined scope, such as exfiltrating data or modifying system configurations beyond the benchmark’s intended “workflow.”

3. **Transparency of Synthetic Persona Construction:** The “PersonaMem-Evo” dataset relies on “implicit behavioral evidence” derived from long interaction histories. The paper mentions using “PersonaHub” and “LLM generators” (Algorithm 3). To ensure ethical compliance, the authors should explicitly state whether the synthetic personas were designed to avoid stereotyping or reinforcing harmful demographic biases, particularly given the inclusion of “anti-stereotypical” preferences. A brief statement on the bias-mitigation strategy during the “CleanAndDeduplicate” step would strengthen the ethical standing of the dataset.

In summary, while the paper identifies high-level risks, it requires specific details on the safety protocols applied during the generation of sensitive synthetic data and the execution of code-based benchmarks to fully address safety and ethics concerns.

Action items.

- [writing] The ‘Broader Impact’ section (App. B) acknowledges privacy risks of persistent memory but lacks a concrete mitigation strategy for the ‘PersonaMem-Evo’ dataset, which contains synthetic health/medical and therapy-related preferences. Explicitly detail the data sanitization protocol or consent simulation used to prevent leakage of sensitive persona traits.
- [writing] The ‘PersonaMem-Evo’ construction (Sec 3.3) utilizes LLMs to generate ‘therapy-related’ and ‘health/medical’ preferences. The paper must clarify if these synthetic personas were subjected to a safety filter to prevent the generation of harmful medical advice or self-harm scenarios, and confirm that no real user data was used to seed these specific categories.
- [writing] The ‘Terminal-Bench-Evo’ and ‘SWE-Chain-Evo’ benchmarks involve agents executing code in evolving environments (Sec 3.1, 3.2). The paper should explicitly state the sandboxing measures (e.g., Docker isolation, network restrictions) used during evaluation to prevent agents from executing unintended dual-use actions (e.g., data exfiltration, system compromise) during the ‘evolution’ phases.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The manuscript presents a novel benchmark (EvoArena) and a memory mechanism (EvoMem) for evaluating LLM agents in dynamic environments. The experimental design covers three distinct domains (terminal, software, social), which strengthens the generalizability of the claims. However, the scientific evidence supporting the efficacy of EvoMem requires further statistical rigor.

The primary concern is the lack of statistical significance testing for the reported performance gains. The paper claims an average accuracy gain of 1.5% on EvoArena, 6.1% on GAIA, and 4.8% on LoCoMo (Introduction; Section 5.2). While these numbers are positive, the manuscript does not provide standard deviations, confidence intervals, or p-values for these differences. Given the relatively small magnitude of the gains (particularly the 1.5% average), it is critical to demonstrate that these improvements are not due to random variance in the evaluation or model stochasticity. For instance, in Table 1, the “Average” step accuracy for Terminal-Bench-Evo increases from 43.6% to 46.0% (+2.4%). Without knowing the variance across the 89 chains or the number of seeds used for evaluation, the robustness of this claim is uncertain.

Additionally, the construction of the PersonaMem-Evo dataset (Section 3.3 and Appendix) relies heavily on LLMs to generate “implicit conversation” histories and “OOD questions.” While the authors mention “dual-blind filtering,” the potential for data leakage or the introduction of generation artifacts remains a threat to validity. If the LLM generator introduces subtle patterns in the conversation history that correlate with the correct answer, agents might exploit these rather than demonstrating true memory evolution tracking. The authors should explicitly describe the metrics or human verification steps used to ensure the generated questions are truly out-of-distribution and not solvable via generation artifacts.

Finally, the sample sizes for the “Average” results in Table 1 are not explicitly defined in the text. The table lists “Average” for Step and Chain accuracy but does not specify if this is an average over the 89 terminal chains, the 48 software chains, or a weighted average across all 1,439 total instances. Clarifying the denominator and the specific subset of models/tasks included in these averages is necessary to interpret the effect sizes correctly.

Action items.

- [science] Report statistical significance (p-values or confidence intervals) for the reported accuracy gains (e.g., +1.5% on EvoArena, +6.1% on GAIA). Without variance estimates or significance testing, it is unclear if these improvements are robust or due to random noise.
- [science] Clarify the sample size and selection method for the ‘Average’ rows in Table

1. Specify the exact number of models and tasks included in the average to assess the stability of the reported 1.5% gain.
- [science] Address potential data leakage in the PersonaMem-Evo construction. The text mentions using LLMs to generate ‘implicit conversation’ and ‘OOD questions’ from seed personas. Detail the filtering process to ensure the test questions do not contain artifacts or patterns from the generation prompts that could be exploited by the agents.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents a novel benchmark and method, but the statistical reporting lacks the rigor required to validate the claimed improvements. While the authors provide point estimates for performance gains (e.g., EvoMem yields a 1.5% average gain on EvoArena, Section 1; Table 1), they fail to report measures of uncertainty such as standard deviations, confidence intervals, or p-values. Without these, it is impossible to determine if the observed improvements are statistically significant or merely artifacts of random variance, especially given the relatively small sample sizes in some subsets (e.g., 10 personas in PersonaMem-Evo).

Furthermore, the paper conducts multiple comparisons across three distinct domains, various agent architectures, and two different metrics (Step Accuracy and Chain Accuracy). There is no mention of a multiple-comparisons correction procedure (e.g., Bonferroni or False Discovery Rate control) to mitigate the increased risk of Type I errors. The “Main Results” section (Section 5.2) and Table 1 present raw differences without context on the variance of the underlying distributions.

Additionally, the definition of the sample size (\$N\$) for the reported averages is ambiguous. For example, in Table 1, the “Average” step accuracy for Terminal-Bench-Evo is listed as 43.6%. It is unclear if this is the mean over the 441 total instances, the 89 chains, or the number of model-agent pairs. The stability of these estimates depends heavily on the effective sample size, which is not explicitly defined in the text or tables. To ensure reproducibility and scientific validity, the authors must provide the standard deviation or standard error for all reported means, specify the exact \$N\$ used for each calculation, and address the statistical significance of the reported gains.

Action items.

- [science] Report statistical significance (p-values or confidence intervals) for the reported performance gains (e.g., -1.5% on EvoArena, -6.1% on GAIA). The current text presents point estimates without indicating if these improvements are statistically distinguishable from noise or baseline variance.

- [science] Clarify the multiple-comparisons correction strategy. With numerous benchmarks (Terminal, SWE, Persona), models, and metrics (Step vs. Chain accuracy), the risk of Type I error is high. Explicitly state if corrections (e.g., Bonferroni, Holm-Bonferroni) were applied or justify the lack thereof.
- [science] Define the unit of analysis and sample size (N) for the reported averages. For instance, in Table 1, is the ‘Average’ step accuracy a mean over 441 instances, 89 chains, or the number of agent-model combinations? The denominator for the percentages must be explicit to assess the stability of the estimates.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a compelling benchmark and method, but the writing quality suffers from several instances of sentence fragmentation and inconsistent phrasing that impede smooth readability.

In Section 3.1 (Executable Workflow Evolution), the text lists dataset statistics in a telegraphic style: “Difficulty: 268 medium, 152 hard, 20 easy, 1 expert.” This lacks a main verb and reads like a raw data dump rather than prose. Similar issues appear in Section 5.1, where the analysis of patch uptake is presented as a conditional fragment (“If patch uptake >0, gain is 8.3%...”) rather than a complete sentence. These fragments disrupt the narrative flow and should be integrated into full sentences.

Furthermore, there is a potential inconsistency in the Introduction versus Section 5.2 regarding the reported average gains. The Introduction cites a “1.5%” average gain on EvoArena, while Section 5.2 specifies “2.6%” for step accuracy and “3.7%” for chain accuracy. While these may refer to different aggregation methods, the lack of explicit distinction in the text creates ambiguity for the reader.

Finally, in Section 5.3, the pronoun “It” in “It improves row-level evidence capture...” is slightly ambiguous without immediate context, though the meaning is recoverable. Tightening these references would enhance precision. Overall, the paper is well-structured, but a pass for grammatical completeness and consistency is required.

Action items.

- [writing] In Section 3.1, the phrase ‘Difficulty: 268 medium, 152 hard, 20 easy, 1 expert’ lacks a verb and reads as a fragment. Rephrase to ‘The dataset difficulty distribution is: 268 medium, 152 hard, 20 easy, and 1 expert task.’ to ensure grammatical completeness.
- [writing] Section 5.1 contains a sentence fragment: ‘If patch uptake >0, gain is 8.3%; if no uptake, gain is 2.6%.’ This should be integrated into a full sentence, e.g., ‘When patch uptake exceeds zero, the performance gain is 8.3%, whereas it drops to 2.6% when no

patches are retrieved.’

- [writing] In Section 5.3, the phrase ‘It improves row-level evidence capture from 72.5% to 74.9%’ is ambiguous regarding the subject. Clarify whether ‘It’ refers to EvoMem generally or the specific mechanism being discussed in that subsection to avoid confusion.
- [writing] The Introduction states ‘EvoMem yields an average gain of 1.5% on EvoArena’ but later in Section 5.2 claims ‘EvoMem improves step accuracy by 2.6% and chain accuracy by 3.7% on average.’ The discrepancy between the 1.5% aggregate and the specific metric gains needs clarification or consistent phrasing to prevent reader confusion.